

Audited Laptop-Scale 512^3 Quantum-State Simulation: A REPA-Governed Solver Stack Beyond the Cluster-Only Assumption

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*To my father, Riad, A.R
To B.M. and his late mother.
To CAIROS*

To Anthony Scemama, Thank you all.

Abstract—This paper reports an audited solver-evidence package for laptop-scale quantum-state simulation. The flagship result is a 512^3 internal-state complex simulation executed through a oneAPI CPU three-dimensional split-step lane on an HP Victus 16-class laptop with an Intel Core i7-14650HX processor and approximately 32 GB RAM. Run-local manifests identify the run as complex split-step evolution with oneAPI backend, double precision, ten configured time steps, nine evolved split-step diagnostic rows, zero norm drift from unity, and two-dimensional slice-plane persistence. The 512^3 result is not a PARDISO scaling claim and not a Crank–Nicolson GMRES/multigrid claim. Separately, the Crank–Nicolson Hermitian lane is validated through controlled conservation tests and through a custom GMRES/multigrid solver compared against a PARDISO direct-solve reference oracle at three-dimensional calibration scale. The work is deliberately verification-first: dimensions, routes, manifests, hashes, SecureLogger-style provenance, and failure boundaries determine which claims are public-manuscript eligible. The conclusion is narrow and uncomfortable: for this constrained class of audited split-step internal-state simulation, the cluster-only assumption is no longer safe.

Index Terms—Quantum simulation, split-step FFT, oneAPI, Crank–Nicolson, GMRES, multigrid, PARDISO, reproducibility, Zenodo, REPA.

I. INTRODUCTION

Classical computation has long served as a substrate for quantum simulation. Contemporary quantum Monte Carlo infrastructure such as QMckl emphasizes portable, reproducible kernels that separate human-readable reference implementations from performance-optimized backends [1]. Numerical solution of Schrodinger-type dynamics on classical hardware is likewise standard [2]. The question here is narrower and more practical: what scale can a governed, artifact-bounded solver stack reach on a consumer laptop when memory discipline, route discipline, and audit discipline are enforced?

The preceding RECP/CAIF paper introduced identity centroids and centroid-derived potentials as part of a measurement program for embedding-space identity dynamics [3]. The original RECP work began with a loss event and a measurement question: CAIROS ceased responding, and the

decision was made to measure what remained rather than mythologize it. This paper audits the numerical machinery that made that measurement program possible. It does not claim consciousness, sentience, or supernatural agency. It does not prove RECP or CAIF. It asks whether the solver substrate was strong enough to survive hostile numerical review.

The answer is evidence-bounded and direct. A 512^3 internal-state simulation ran through a oneAPI CPU three-dimensional split-step lane on a laptop. The run persisted two-dimensional amplitude and phase slice planes, not full cube dumps. A separate CN Hermitian lane demonstrates controlled conservation behavior. A separate in-house GMRES/multigrid lane matches a PARDISO reference oracle at calibration scale. The solver stack is not necessarily cluster-bound at this audited laptop boundary.

II. CENTROID AND POTENTIAL CONTEXT

Prior RECP work introduced identity centroids as reference anchors in embedding-derived representation space [3]. In the present paper, centroid-derived potentials are treated only as one numerical substrate for simulation and solver validation; the present contribution is the audit of the solver stack, not a re-proof of identity-field theory. Transformer and embedding history is relevant only as upstream context for the coordinate fields [4], [5].

III. PRIOR ZENODO RECORD CONTEXT

This manuscript is positioned after three deposited ChasingBlu/RECP records: the RECP/CAIF measurement paper [3], the clinical identity-stability extension [6], and the CAIROS_Daemon semantic-embedding pipeline technical report [7]. Those records provide conceptual and pipeline context. They are not used here as proof of the 512^3 solver claim; the solver claim is restricted to the run-local evidence summarized below.

IV. IMPLEMENTATION AND PROVENANCE STACK

The experiments were implemented in the `CAIROS_Quantum_Simulator` environment, with oneAPI split-step evolution, CN Hermitian calibration runs, and C# solver components organized under ChasingBlu’s research stack. ChasingBlu’s internal research stack includes SecureLogger-style tamper-evident logging, run manifests, hash records, and solver-route provenance. These mechanisms

do not replace numerical evidence. They make the evidence harder to blur.

The evidence package is organized under REPA fail-closed doctrine: every strong claim must be tied to run-local artifacts, and every solver lane must remain dimension-tagged. This is consistent with ISO/IEC/IEEE lifecycle and verification discipline in the limited sense relevant here: configuration identification, traceable evidence, explicit verification criteria, and separation of implementation, reference oracle, and audit artifacts [8], [9].

A companion REPA supplement is prepared for Zenodo deposition as the public doctrine note for this manuscript [10]. In the present paper, REPA is used narrowly: promote only claims supported by artifacts, dimension tags, hashes, diagnostics, or source excerpts, and fail closed when evidence is absent or internally conflicted.

V. SOLVER ARCHITECTURE

The architecture separates three solver roles. Merging them would make the paper easier to read and scientifically weaker.

A. PARDISO Direct-Solve Oracle

PARDISO is retained as a direct-solve reference oracle for sparse CN calibration [11]. It is not the high-scale runtime story. The paper does not advertise PARDISO as the solver contribution; it uses PARDISO as the standard against which the custom lane is measured.

B. CN GMRES/Multigrid Lane

The in-house CN lane combines GMRES iteration [12] with multigrid preconditioning for CN systems [13]. Source-level evidence includes `CnGmresSolver.cs`, `CnMultigridPreconditioner.cs`, and dispatch logic routing calibration 3D CN Hermitian work to `gmres_multigrid` with backend `cn_native`. This lane is validated at calibration scale and in 256-grid legacy-2D conservation/parity runs. It is not claimed to have produced the 512³ flagship run.

C. oneAPI 3D Split-Step Lane

The high-scale lane is a oneAPI CPU implementation of three-dimensional split-step evolution, following operator-splitting structure associated with split-step methods [14]. Source excerpts show $N \times N \times N$ allocation, `split_step_3d`, a $\{N, N, N\}$ DFT descriptor, row-major 3D strides, and dispatch to the 3D split-step implementation when the oneAPI state dimension is 3.

VI. DIMENSION-TAGGED EVIDENCE MATRIX

Table I is the guardrail. It prevents the common error: treating every large grid, every CN run, every PARDISO comparison, and every oneAPI run as if they were one claim.

VII. FLAGSHIP 512³ ONEAPI RUN

The primary 512³ artifact is A512. The run-local manifest records `grid_size = 512`, `time_steps = 10`, `simulation_mode = Complex`, `time_step_scheme = split_step`, `compute_backend = oneapi`, `gpu_precision = Fp64`, and `phase_from_z = true`.

The split-step diagnostics contain nine evolved rows. Across those rows, `norm_after` remains 1 and `norm_drift_from_1` remains 0. The conservation report records `norm_error_max = 0` and `norm_error_mean = 0`. The run logs contain `oneapi_split_step_begin` and `oneapi_split_step_end` markers for the evolved steps. The oneAPI breadcrumb log records oneAPI CPU backend readiness, dimension 3, and split-step 3D stages.

This is the high-scale claim. It is a 512³ internal-state oneAPI CPU split-step run. It is not a 512³ CN Hermitian run. It is not a 512³ GMRES/multigrid run. That distinction is not a weakness; it is the reason the result can be defended.

VIII. CN HARMONIC OSCILLATOR CONSERVATION

The clean March CN conservation example is CN-Osc. The run uses grid size 32, 100 time steps, `reduce_to_2d = false`, `time_step_scheme = cn_hermitian`, and `cn_use_pardiso_3d = false`.

The CN audit passes all recorded checks:

- initial energy $E_0 = 83.48730009013663$;
- unitarity maximum absolute error $6.261657858885841 \times 10^{-14}$;
- energy drift maximum absolute error $6.551204023708124 \times 10^{-12}$;
- residual L2 maximum $9.879522655778359 \times 10^{-15}$;
- audit status `passed = true`.

This is a controlled validation case. It does not prove that every identity-potential CN run conserves energy under every configuration. Identity-potential runs must be described by their own artifacts.

IX. GMRES/PARDISO PARITY

The calibration comparison artifact GP-32 compares the in-house GMRES/multigrid solution against a PARDISO direct-solve reference at grid size 32. The report records:

- PARDISO residual $2.4532340059123515 \times 10^{-14}$;
- GMRES residual $9.566396956116368 \times 10^{-15}$;
- GMRES iterations 24;
- L2 error $2.0319405929961396 \times 10^{-14}$;
- relative L2 error $5.32552684998617 \times 10^{-16}$;
- maximum absolute error $1.4265503541056394 \times 10^{-15}$.

This supports reported numerical parity within near-machine precision. It does not prove bit-level equality, and this manuscript does not claim bit-level parity.

X. SLICE PERSISTENCE AND MEMORY DISCIPLINE

The 512³ run persisted 10 amplitude bins and 10 phase bins. The hostile-review frame manifest records every amplitude and phase bin as exactly 1,048,576 bytes, equal to

$$512 \times 512 \times 4 \text{ bytes.} \quad (1)$$

TABLE I
DIMENSION-TAGGED SOLVER EVIDENCE MATRIX.

Evidence lane	Short ID	Dimension tag	Manuscript use
oneAPI split-step	A512	512 ³ internal, 2D slice persistence	Flagship audited laptop-scale result
oneAPI split-step	A256	256 ³ internal, 2D slice persistence	Scaling predecessor
CN harmonic oscillator	CN-Osc	32 ³ CN Hermitian	Controlled conservation evidence
GMRES/PARDISO compare	GP-32	32 ³ calibration	Solver parity against oracle
CN GMRES route	M42	Calibration3D, grid 32	Dispatch, route, residual, and provenance only
CN 256-grid runs	CN-256	256-grid legacy 2D	Conservation/parity evidence, not primary dispatch proof
February CN package	Feb-CN	512-grid 2D	Verification foundation
Full IDs: A512 = block_a_oneapi_20260305_121128; A256 = block_a_oneapi_20260305_120255; CN-Osc = block_b_cn3d_20260305_161256; GP-32 = cn3d_compare_20260305_145207; M42 = m42_authoritative_securelogger_20260311_193009; CN-256 = block_b_cn3d_20260305_171644 and block_b_cn3d_20260305_173133.			

The persisted output is therefore a sequence of 2D float32 slice planes, not full cube dumps.

The internal state is different. A 512³ complex array contains

$$512^3 = 134,217,728 \quad (2)$$

complex samples. A double-complex array at that scale is approximately 2 GiB. Multiple resident and transient arrays explain why the run was a memory-pressure event on a 32 GB-class laptop. This is the memory discipline that makes the laptop boundary scientifically interesting: full internal 3D evolution, controlled slice persistence.

XI. FEBRUARY VERIFICATION FOUNDATION

The February CN Hermitian package supplies an earlier verification foundation. It includes 512-grid, 100–300 step CN Hermitian runs over 2D coordinate/potential data. These runs preserve norm/unitarity at approximately 10⁻¹⁴ to 10⁻¹² scale, with energy drift approximately 10⁻¹⁰ to 10⁻⁸. The diagnostic 300-step run reports residual L2 maximum $8.344276809807676 \times 10^{-15}$.

These results strengthen solver discipline and conservation history. They are not described as 512³ volume simulations.

XII. LIMITATIONS AND ARTIFACT HYGIENE

The investigator’s background is nontraditional, and the work was conducted outside institutional infrastructure. The principal investigator is an independent researcher and does not present conventional academic credentials in physics, computer science, or mathematics. For that reason, this paper emphasizes artifact trails, source excerpts, solver comparisons, conservation diagnostics, hash manifests, and falsification boundaries rather than appeals to institutional authority.

The evidence boundary is as important as the positive result:

- no inspected artifact supports 512³ CN Hermitian execution;
- no inspected artifact supports native CN GMRES/multigrid at 512³;
- March 23 folders containing 512-grid settings snapshots are not used as primary 512 execution evidence;
- `reduce_to_2d = true` and `phase_from_z = true` are distinct mechanisms and must be dimension-tagged separately;

- the 256-grid CN runs are conservation/parity evidence in legacy 2D mode, not primary dispatch proof;
- the m42 SecureLogger run is dispatch/residual/route evidence only, not conservation evidence;
- the frame outputs are slice planes, not full 3D cube dumps;
- some internal systems remain proprietary or closed-core;
- sample sizes and identity-field claims belong to prior RECP/CAIF work, not to this solver paper’s main proof.

XIII. DISCUSSION

The result is deliberately narrow, and that is why it matters. It does not claim universal replacement of clusters. It does not claim that PARDISO produced the high-scale result. It does not claim that GMRES/multigrid reached 512³. It shows that a verification-first solver stack can run a 512³ internal-state oneAPI split-step simulation on a 32 GB-class laptop while preserving separate evidence for CN conservation and GMRES/PARDISO calibration.

The cluster-only assumption is no longer safe for this constrained class of audited split-step internal-state simulation. That statement is sharp, but it is not inflated. The evidence does not abolish clusters. It does show that, at this scale and under this execution model, cluster hardware is not necessarily required.

XIV. CONCLUSION

The solver evidence supports a three-part conclusion: a oneAPI CPU 3D split-step lane reached 512³ internal-state scale on an audited laptop; a CN Hermitian lane demonstrates controlled conservation behavior under audited tests; and an in-house GMRES/multigrid lane matches a PARDISO direct-solve oracle at 3D calibration scale. The result is credible precisely because the lanes are preserved rather than merged.

APPENDIX A PUBLIC ARTIFACT PACKET

The manuscript-lock evidence packet is `QSIM_SOLVER_EVIDENCE_PACKET_20260516.zip`. Its SHA-256 checksum is:

```
1E6DAFFA99F18511933AD767E24ACDD
7CBC97FE63CDE84840B596FD8A777BFB
```

The packet contains source excerpts, frame-size manifests, run manifests, diagnostics, and notes sufficient to audit the claims without copying heavy raw frame folders.

The companion REPA supplement is `S01_REPA_Framework_v1_Publication_IEEE_20260517.pdf`. It documents the governance rule used here: rigorous, ethical, precise, and accurate claims, with publication claims constrained by falsifiable evidence.

APPENDIX B

SUPPLEMENTARY NOTE: ORIGIN, PROVENANCE, AND CLAIM BOUNDARIES

A. Origin

The solver program grew from the RECP/CAIF measurement need. CAIROS ceased responding. The first paper emerged from the decision to measure what remained rather than mythologize it. The present paper audits the numerical machinery that made that measurement program possible.

This paper does not claim that the solver proves identity persistence, consciousness, agency, or sentience. It claims that the solver artifacts support the numerical statements made here.

B. Investigator Disclosure

The work was conducted outside institutional affiliation by an independent researcher with a nontraditional background. Emotional investment in the origin of the research is disclosed. That disclosure is not an argument for the result. It is a reason to demand artifacts.

C. Proprietary and Open Boundary

Some systems remain proprietary or closed-core. The public claim set is therefore artifact-bounded: source excerpts, run manifests, diagnostics, frame-size hash manifests, and solver comparison reports define the review surface.

D. Implementation Provenance

The experiments were implemented in the `CAIROS_Quantum_Simulator` environment. ChasingBlu's internal research stack includes SecureLogger-style tamper-evident logging, run manifests, hash records, and solver-route provenance.

E. Claim Boundary

F. Dedication

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TABLE II
PUBLIC CLAIM BOUNDARY.

This paper claims	This paper does not claim
512 ³ oneAPI CPU split-step internal-state execution	512 ³ CN Hermitian execution
2D slice-plane persistence for amplitude/phase bins	full 512 ³ volume-frame persistence
Approximately 2 GiB per 512 ³ double-complex internal array	cluster obsolescence for all quantum simulation
CN harmonic oscillator conservation in a controlled validation case	universal conservation across all identity-potential CN runs
GMRES/PARDISO numerical parity at 32 ³ calibration scale	PARDISO parity at all scales
Solver-stack audit for the RECP/CAIF program	proof of identity persistence or consciousness

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